

Operators in list

1. Arithmetic:-

```
l1 = [1,2,33]
l2 = [1,2,3]
print(l1+l2)
print(l1*3)

# Output
[1, 2, 33, 1, 2, 3]
[1, 2, 33, 1, 2, 33, 1, 2, 33]
```

2. Assignment:-

```
l1 = [1,2,33]
l1+=l2
print(l1)
l1*=5
print(l1)

# Output
[1, 2, 33, 1, 2, 3]
[1, 2, 33, 1, 2, 3, 1, 2, 33, 1, 2, 3, 1, 2, 33, 1, 2, 3, 1, 2, 33, 1, 2, 3, 1, 2, 33, 1, 2, 3]
```

3. Comparision:-

```
l1 = [1,2,33]
l2 = [1,2,3]
print(l1==l2)
print(l1!=l2)
print(l1<l2)
print(l1<=l2)
print(l1>l2)
print(l1>l2)
```

```
# Output
False
True
False
False
```

```
True
```

```
True
```

4. Identity:-

```
l1 = [1,2,3,4]
l2 = [1,2,3,4]
print(l1 is l2)
print(l1 is not l2)
```

```
# Output
```

```
False
```

```
True
```

5. Membership:-

```
l1 = [1,2,3,4]
l2 = [1,2,3,4]
print(l1 in l2)
print(l1 not in l2)
```

```
# Output
```

```
False
```

```
True
```

6. logical:-

```
# Non-empty list → True
# Empty list " " → False
## and
```

```
l1 = [1,2,3,4]
l2 = [1,2,3,5]
l3 = []
print(l3 and l3)
print(l3 and l2)
print(l1 and l3)
print(l1 and l2)
print(l2 and l1)
```

```
# Output
```

```
[]
```

```
[]
```

```
[]
```

```
[1, 2, 3, 5]
```

```
[1, 2, 3, 4]
```

```
## or
```

```
l1 = [1,2,3,4]
l2 = [1,2,3,5]
l3 = []
print(l3 or l3)
print(l3 or l2)
print(l1 or l3)
print(l1 or l2)
print(l2 or l1)
```

Output

```
[]
[1, 2, 3, 5]
[1, 2, 3, 4]
[1, 2, 3, 4]
[1, 2, 3, 5]
```

not

```
l1 = [1,2,3,4]
l2 = []
print(not l1)
print(not l2)
```

Output

```
False
True
```